Homework 1
FIN2020 Fall 2023

Due **18:00 Friday, Sep. 22 2023 Beijing Time**

Problem 1

Solve the constrained optimization problem:

$$\max_{c_0,c_1,c_2} \quad ln(c_0) + ln(c_1) + ln(c_2) \tag{1}$$

$$\text{subject to } p_0c_0 + p_1c_1 + p_2c_2 \leq Y, \tag{2}$$

where ln is the natural logarithm function.

a. Set up the Lagrangian for the problem. Then write down the first order conditions for choice variables.

b. Will the constraint bind or not? Why.

c. How the optimal solution depends on ps and Y.

d. If we interpret this problem as consumer's optimization problem as follows. A consumer consumes c_0, c_1 and c_2 units of good 0, 1, and 2 in a perfectly competitive market at the unit price p_0, p_1 and p_2. The consumer's objective can be written as Equation (1), and the consumer is subject to the budget constraint of Equation (2), where Y is the total income of the consumer. Then what fraction of the consumer's total income Y does the consumer spend on each good?

Problem 2

Solve the constrained optimization problem:

$$\max_{y,k,l} \quad py - rk - wl \tag{3}$$

$$\text{subject to } k^a l^b \geq y, \tag{4}$$

where $0 < a < 1$, $0 < b < 1$, and $0 < a + b < 1$, and take p, r and w as given.

a. Set up the Lagrangian for this problem, letting λ denote the multiplier on the constraint.

b. Next, write down the conditions that, according to the Kuhn-Tucker theorem, must be satisfied by the values y^*, k^*, and l^* that solve the firm's problem, together with the associated value λ^* for the multiplier.

c. Assume that the constraint binds at the optimum (can you tell under what conditions this will be true?), and use your results from above to solve for y^*, k^*, l^*, and λ^* in terms of the model's parameters: a, b, p, r, and w.

d. If we interpret y as the product produced by a firm, k as the capital and l as labor used by the firm. Then p can be viewed as the unit price of product sold by the firm, r can be viewed as the rents of each unit of capital, and w can be viewed as the wage for each unit of labor. Then use your solutions from above to answer the following questions:

   i. What happens to the optimal y^*, k^*, and l^* when the product price p rises, holding all other parameters fixed? In each case, does the optimal choice rise, fall, or stay the same?

   ii. What happens to the optimal y^*, k^*, and l^* when the rental rate for capital r rises, holding all other parameters fixed?

   iii. What happens to the optimal y^*, k^*, and l^* when the wage rate w rises, holding all other parameters fixed?

   iv. What happens to the optimal y^*, k^*, and l^* when p, r, and w all double at the same time?

Problem 3

Consider a Gaussian random variable X, i.e., X follows a normal distribution with mean μ and variance σ^2:

$$X \sim \mathcal{N}(\mu, \sigma^2). \tag{5}$$

There is a function of X:

$$u(X) = -e^{-\nu X}, \tag{6}$$

where $\nu > 0$ is a parameter.

a. Compare $\mathrm{E}[u(X)]$ and $u(\mathrm{E}[X])$. Are they equal? If not, which one is bigger.

b. Compute $\mathrm{E}[u(aX)]$, where a is a choice variable.

c. Show that the optimization problem

$$\max_a \mathrm{E}[u(aX)] \tag{7}$$

is the same as the following one

$$\max_a \quad \mu a - \frac{1}{2}\nu\sigma^2 a^2. \tag{8}$$

Hint: The probability density function for normal distribution with mean μ and variance σ^2 is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

The expected value of a function $g(x)$, when x is a random variable with a probability density function $f(x)$, is $\int_{-\infty}^{\infty} g(x)f(x)dx$.